AIAC PROJECT

Student ID card detection system using YOLOv8

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Prompts we used in the project :

Step 1: Install Libraries

Prompt: "Install the necessary libraries for a YOLOv8-based detection system, including ultralytics, opencv-python-headless, and numpy."

Step 2: Import Modules

Prompt: "Import the required Python modules such as cv2, numpy, YOLO from ultralytics, deque, matplotlib.pyplot, and IPython.display utilities."

Step 3: Load Person Detection Model

Prompt: "Load a pre-trained YOLOv8 model (yolov8n.pt) specifically for detecting persons."

Step 4: Load Custom ID Card Model (or Demo Mode)

Prompt: "Create a section to load a custom ID card detection model (best.pt) from an uploaded file, but also include a demo mode option if no custom model is used, which simulates ID card boxes."

Step 5: Define Helper Function

Prompt: "Define several helper functions: one to calculate Intersection Over Union (IoU), another to check if a person's bounding box overlaps with an ID card box, a StabilityBuffer class to ensure consistent detection, and a function to draw bounding boxes and labels on an image."

Step 6: Load Image

Prompt: "Add code to allow the user to upload an image from their local machine and load it using OpenCV."

Step 7: Perform Detection and Visualization

Prompt: "Implement the main detection logic: perform person detection, then ID card detection (or simulation), determine which persons are compliant, draw the results with bounding boxes and labels, include a summary of counts, and finally display the output image using matplotlib."